MLOps CLA 4: ML Model Deployment and Model Serving

Dataset: Kaggle Titanic — Machine Learning from Disaster (train.csv)

Objective

Train and save a classification pipeline, demonstrate online, batch and streaming inference, and serve predictions using FastAPI.

Model and evaluation

Classifier: Random Forest with preprocessing for missing numeric and categorical values. The model is evaluated on a held-out 20% split. Measured holdout accuracy: 0.8156.

Observations

Question	Observation
Online vs Batch	Online inference predicts one request at a time; batch inference predicts a collected group of rec
Streaming vs Batch	Streaming handles incoming records continuously with low latency; batch processes a group aft
Why FastAPI?	It provides HTTP endpoints, Pydantic request validation, and interactive API documentation at /d
Why reuse one model?	It keeps inference consistent and reproducible across online, batch, streaming and API serving.

Submission note

Attach screenshots showing online, batch and streaming outputs, successful /health and /predict tests, and one invalid request rejected by validation. The inference records are selected from the supplied Kaggle training file with their labels excluded from prediction input; they are demonstration records, not an external unlabeled dataset.